

**Rethinking Underground Space Utilization in Metro Stations:
A Case Study of Shanghai Jing'an Temple**

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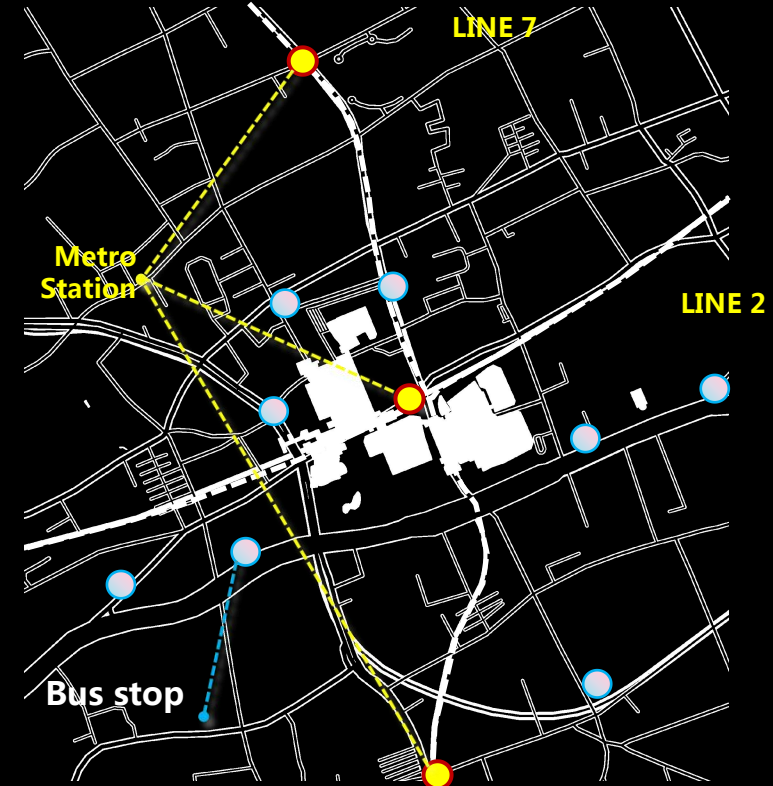
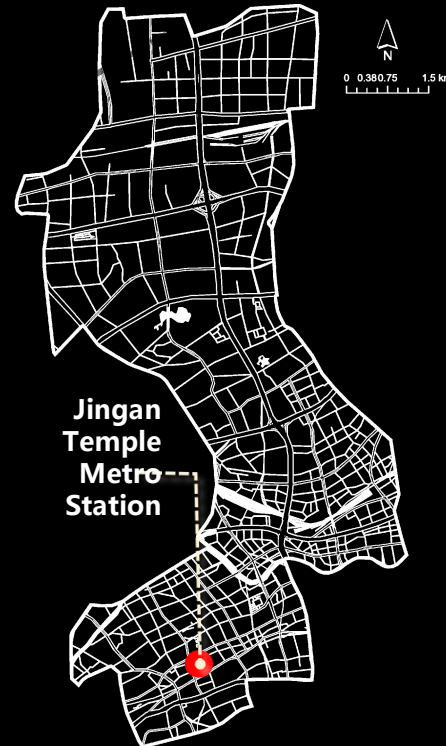
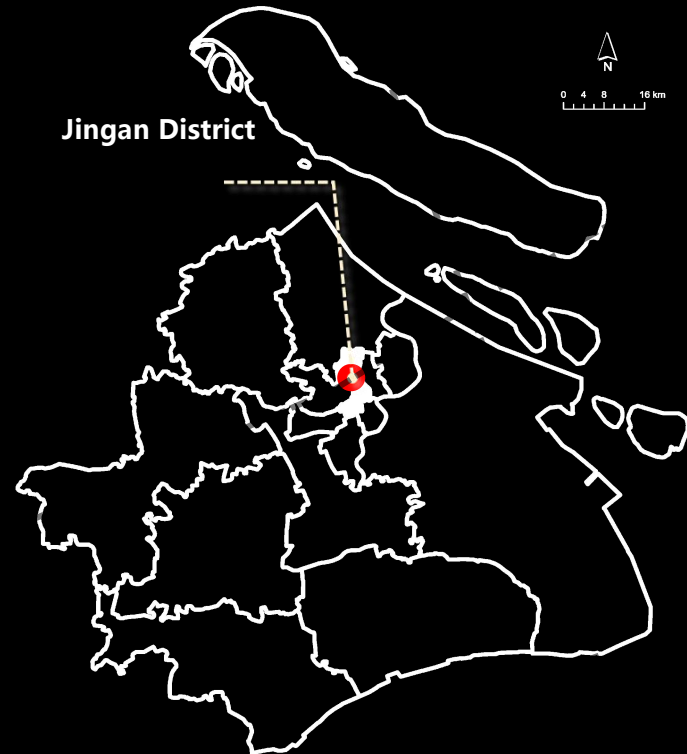
1. Research Background

With the rapid development of high-density cities, **urban underground space (UUS)** has become **an important component of urban spatial systems**. **Metro stations**, as key public transport hubs, often sit at the core of underground space networks and provide strong potential for functional integration and efficient use of urban land

□ Research Objective

Measure the **spatial** characteristics of UUS around **high-density urban transit hubs**
Explore strategies for **improving spatial vitality and efficiency**

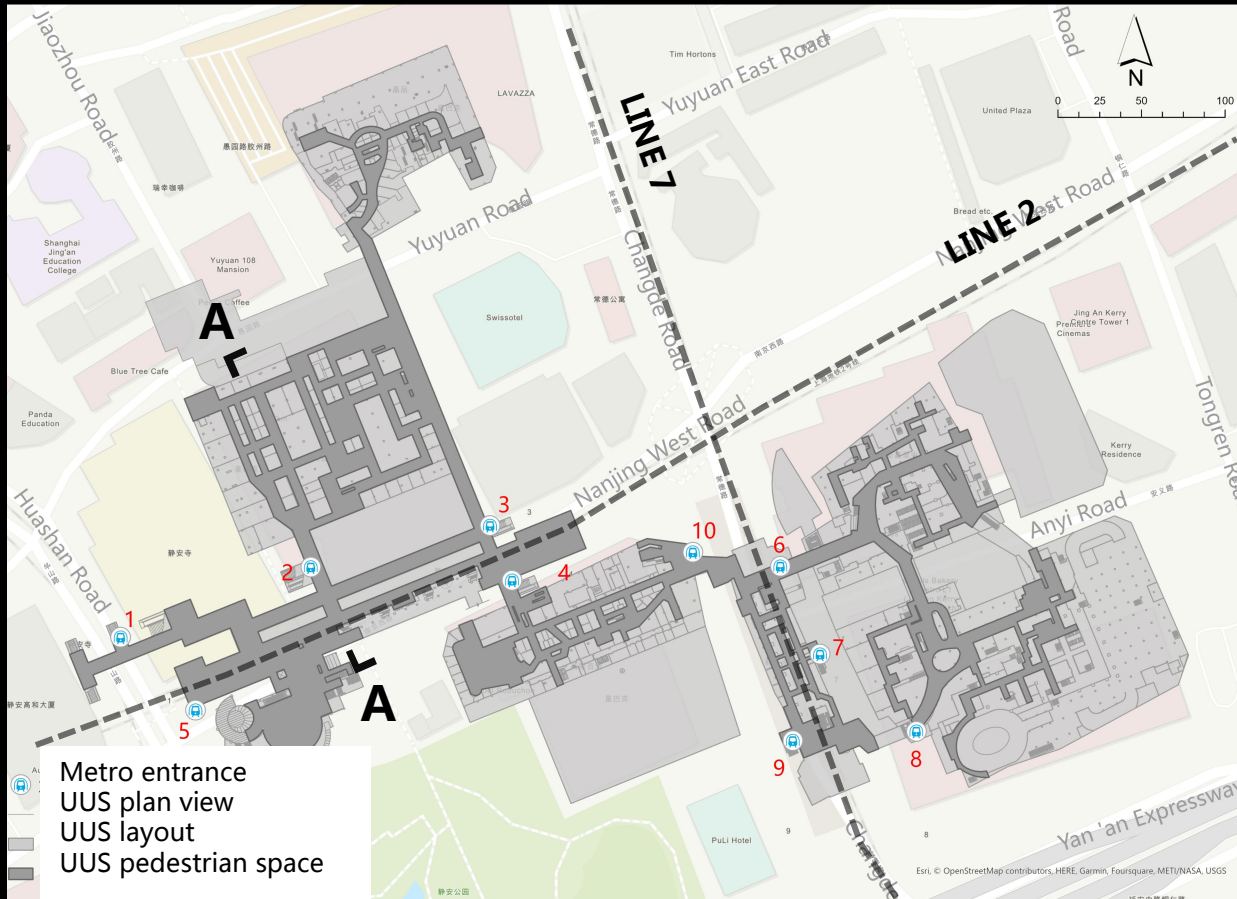
□ Case Study Site: **Shanghai Jing' an Temple Metro Station**.



The Location of Study Area

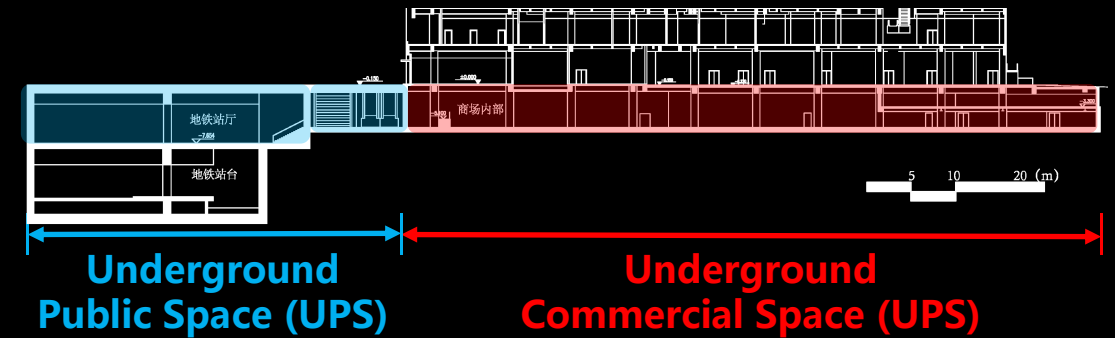
1. Research Background

Analysis is based on **two levels of spatial scope**



The site of UUS of **Shanghai Jing 'an Temple (SJTUUS)**

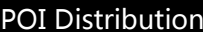
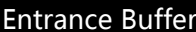
- ① **External space**: area surrounding the metro station (location, accessibility, land use, POI distribution).
- ② **Internal space**: divides into
 - **UPS (Underground Public Space)** — underground corridors, underground commercial street, concourse.
 - **UCS (Underground Commercial Space)** — underground commercial areas of nearby malls connected to the station



A-A Section of Jing 'an Temple subway station

1. External Spatial Characteristic

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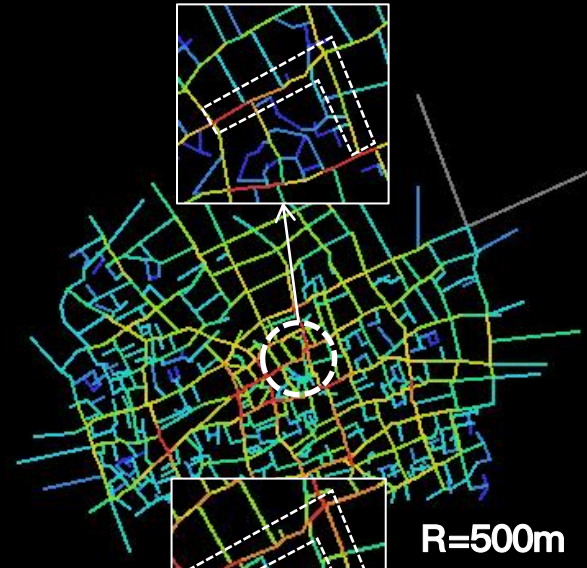
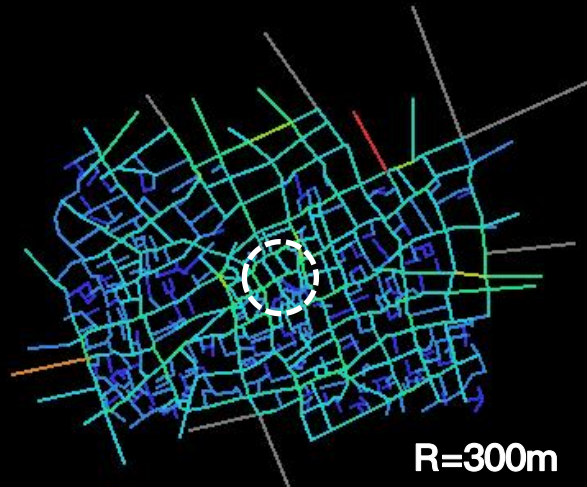


- ❑ **Land-use patterns** : directly connected to Jiuguang Department Store, Crystal Galleria, Réel Mall, Jing' an Park, and Kerry Centre, showing a **highly concentrated commercial development pattern** around the transit hub.
- ❑ **POI distribution** : Using the 10 station entrances as centers, 14 categories of POI data were collected within a 750 m radius. Results indicate : a **high concentration of POIs related to daily services, dining, and retail.** These facilities are **scattered along streets but cluster around the metro station**

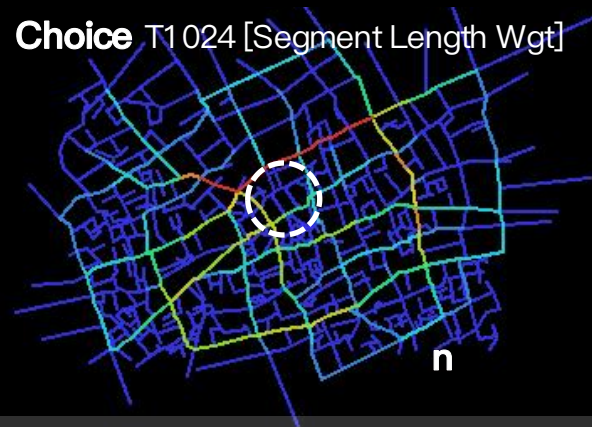
2. External Spatial Characteristics of SJTUUS

Space Syntax——Segment Analysis

Integration T1024 [Segment Length Wgt]



Choice T1024 [Segment Length Wgt]



- From both the global and local perspectives, the **integration and choice values** of the road network around the metro station are **relatively high**, indicating that this area functions as a **central zone** within its surrounding network.
- Within the commonly used **5-, 10-, and 15-minute walking ranges**, the **spatial centrality** of the area becomes **more pronounced** as the radius increases, demonstrating **higher accessibility** and is **more likely to be used by pedestrians**.

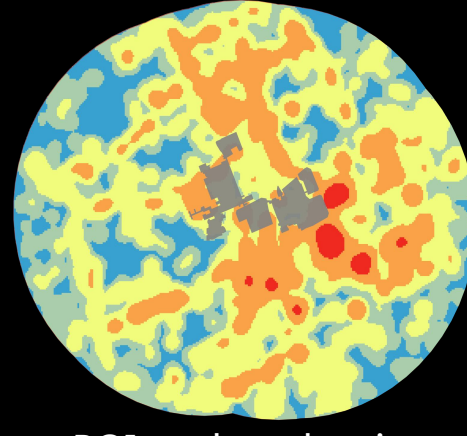
2. External Spatial Characteristics of SJTUUS



Integration (Rn)



Integration (R500)



POI nuclear density



Choice (Rn)



Choice (R500)



POI diversity (Shannon Entropy)

□ Overall:

Nanjing West Road and Yan' an Middle Road, located near Jing' an Temple Station, are key east–west corridors with high accessibility and strong choice values. These streets also host a large number and wide variety of facilities, making them major activity axes in the area.

□ TOD Feature Matching:

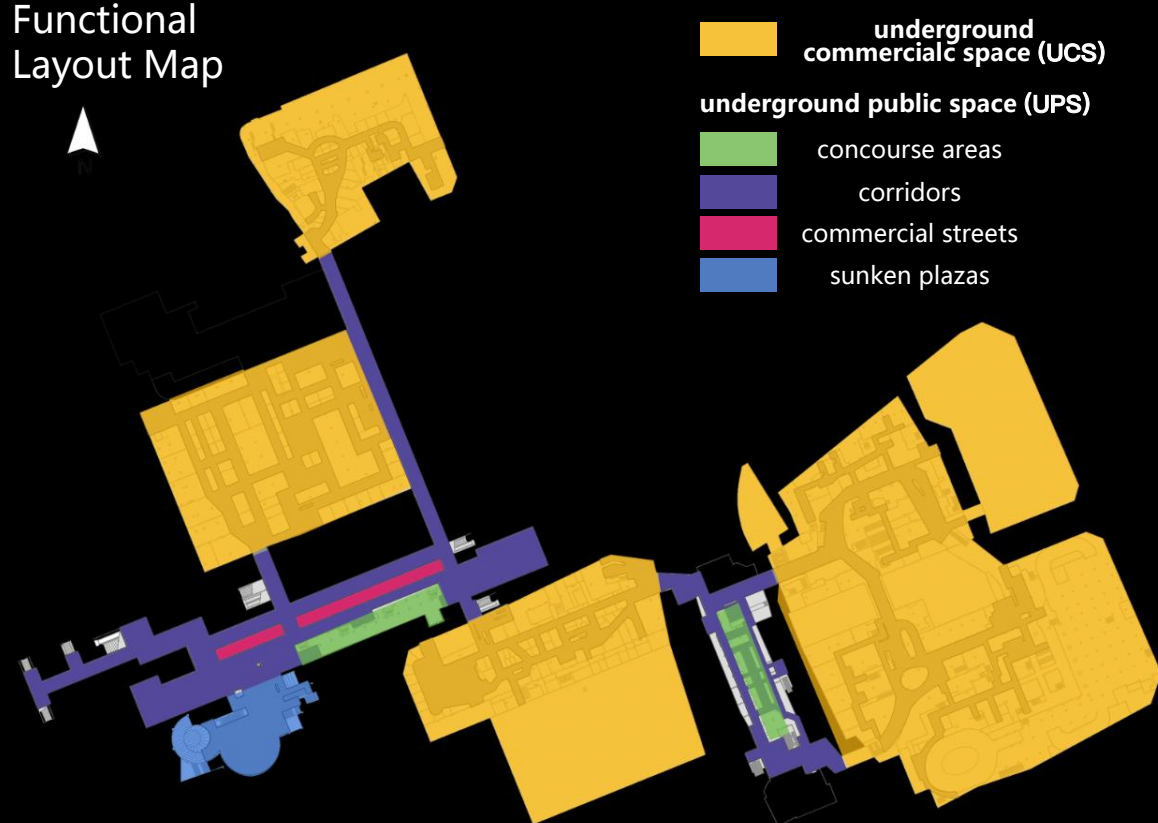
The spatial analysis results align with the diversity and kernel density distribution of POI facilities around the station.

The development pattern meets Shanghai' s regulatory planning requirement of increasing development intensity within 500 m of metro stations.

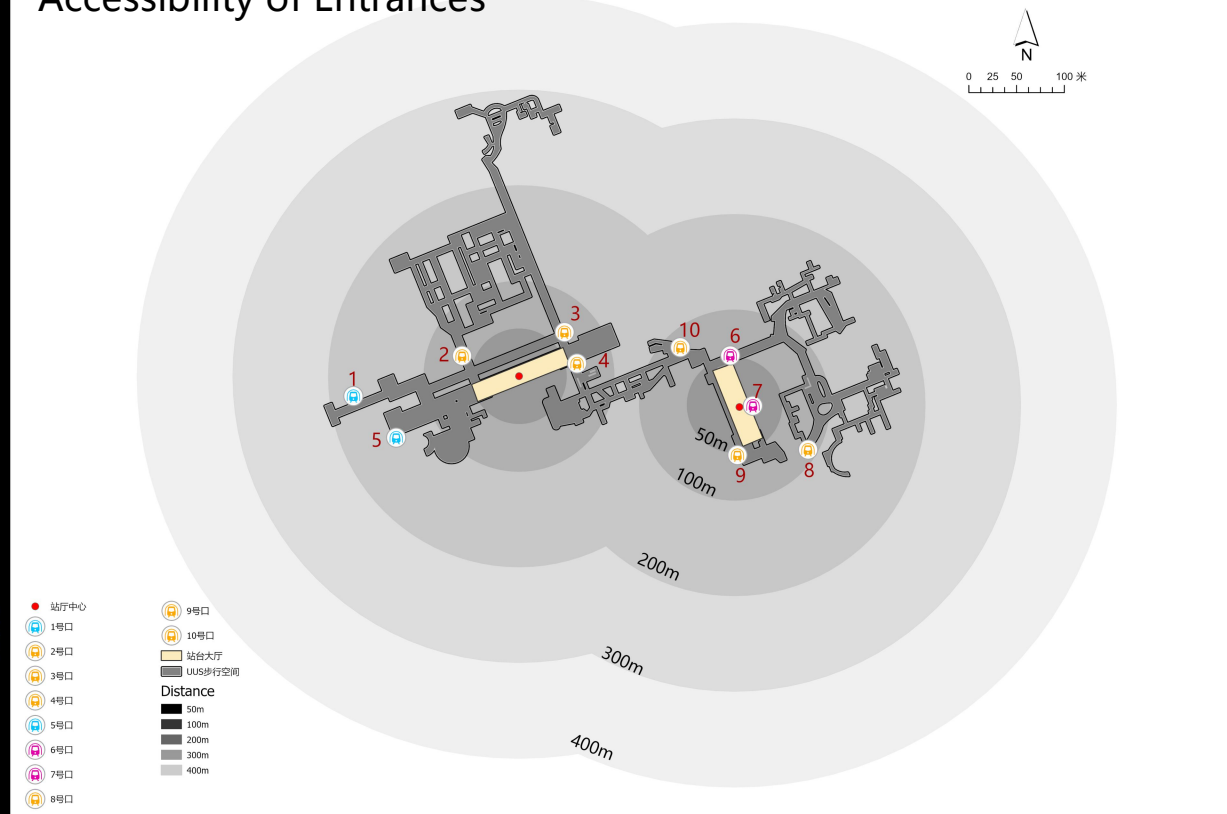
2. Interior Space Characteristic

3. Interior Space Characteristics of SJTUUS

Functional
Layout Map

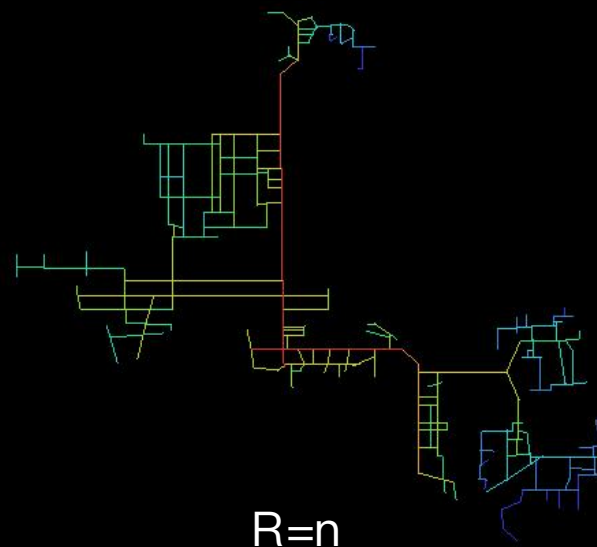
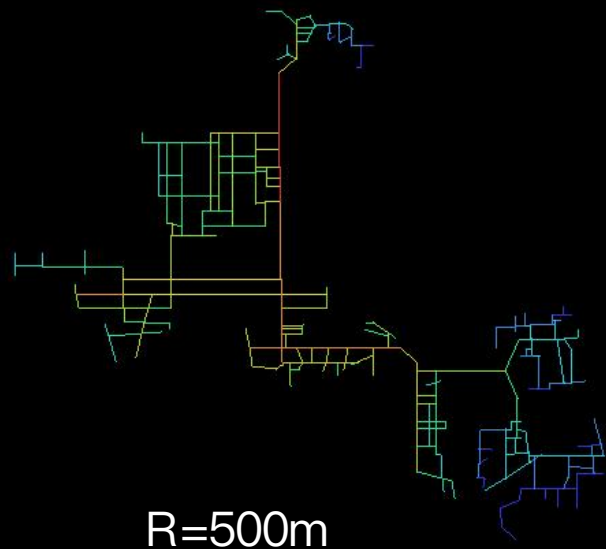
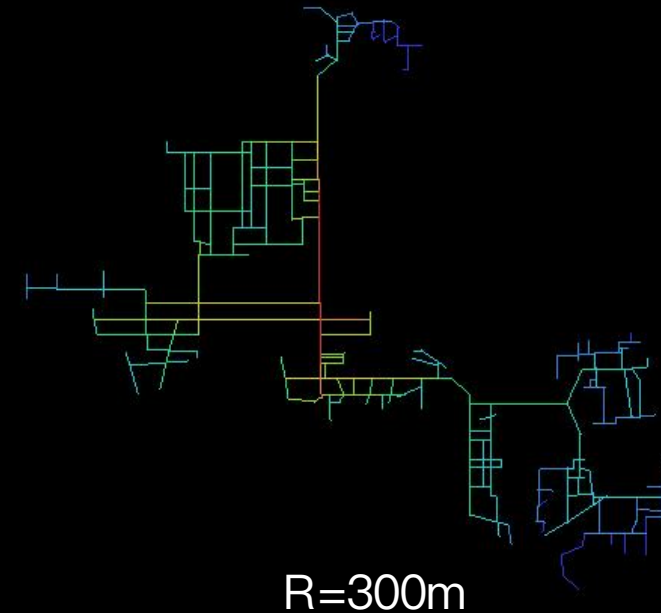
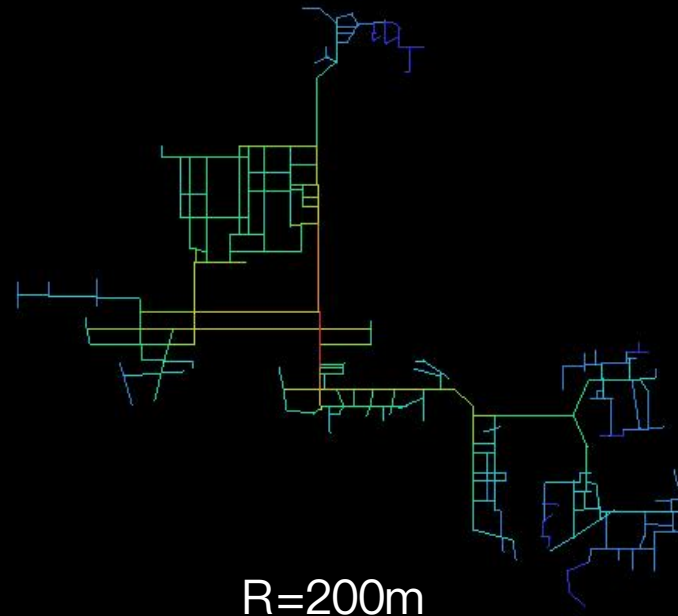
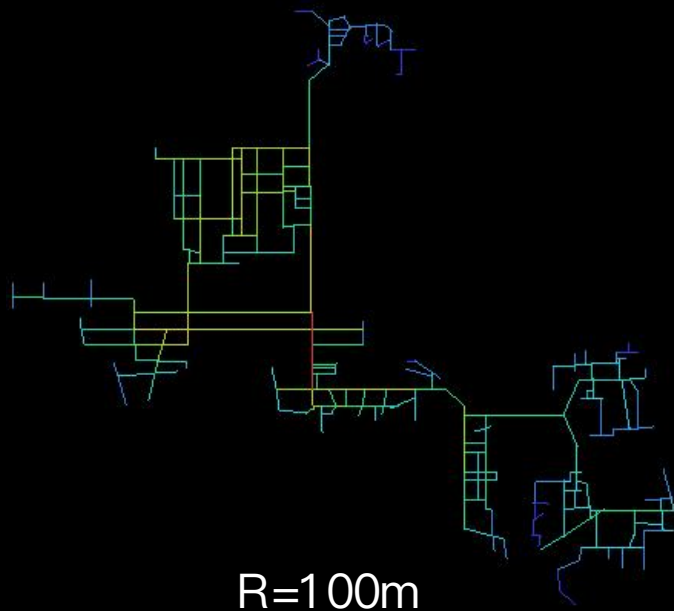


Accessibility of Entrances



- The **underground public space (UPS)** of this station includes **concourse areas, corridors, commercial streets, and sunken plazas**, while the **underground commercial space (UCS)** is mainly distributed around the perimeter.
- The SJTUUS as a whole integrates **transportation, commerce, leisure, and circulation functions**, with strong social service capacity.
- Results show that **80% of the entrances are within a 100 m walking distance of the concourse**. Entrance No. 7 has the highest accessibility, directly linking to Crystal Galleria, while accessibility near Crystal Galleria is relatively lower.

3. Interior Space Characteristics of SJTUUS



Segment Analysis

Integration T1024 [Segment Length Wgt]

Integration Characteristics:

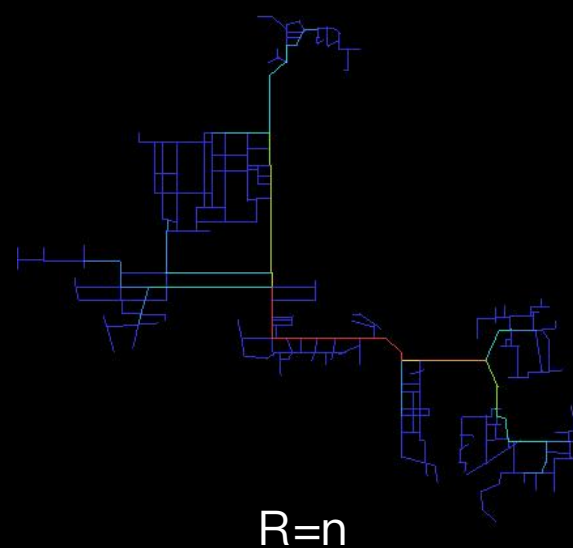
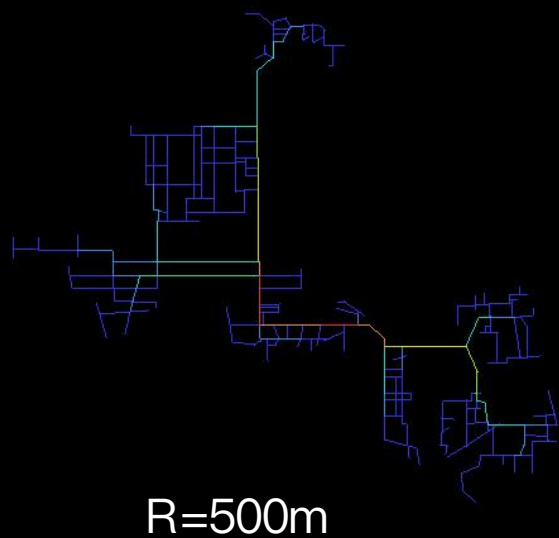
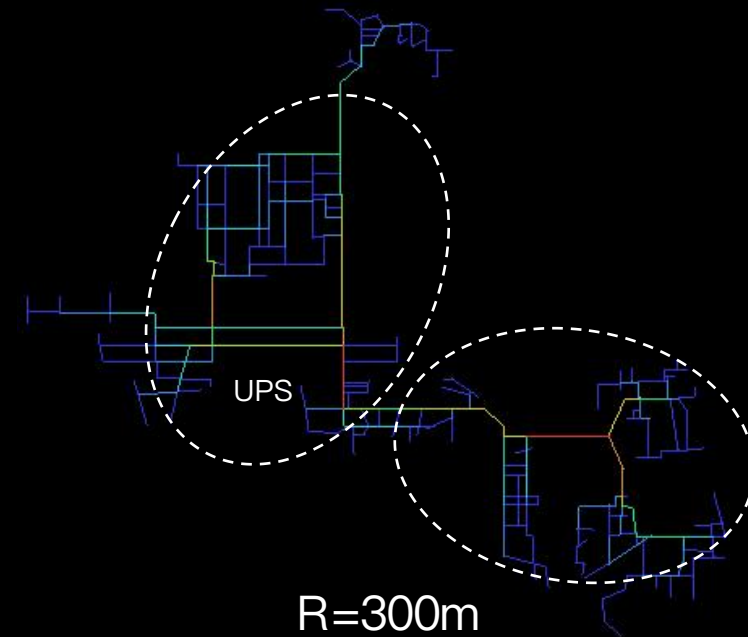
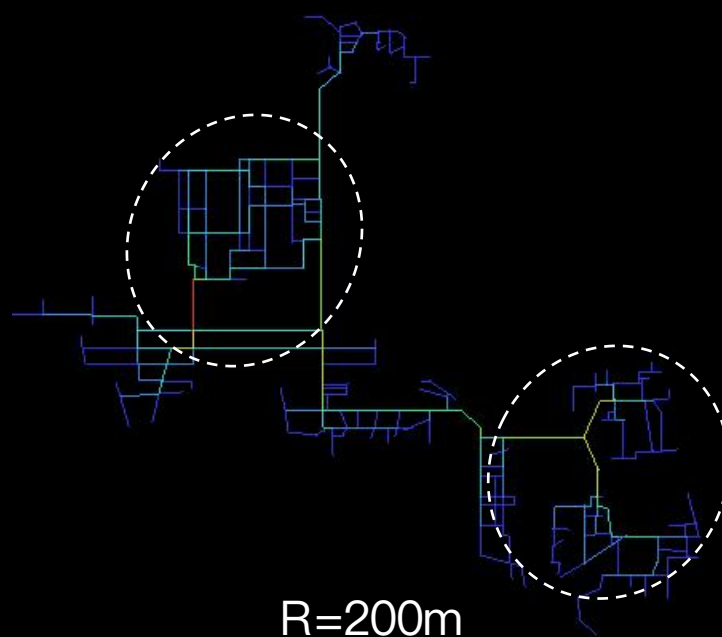
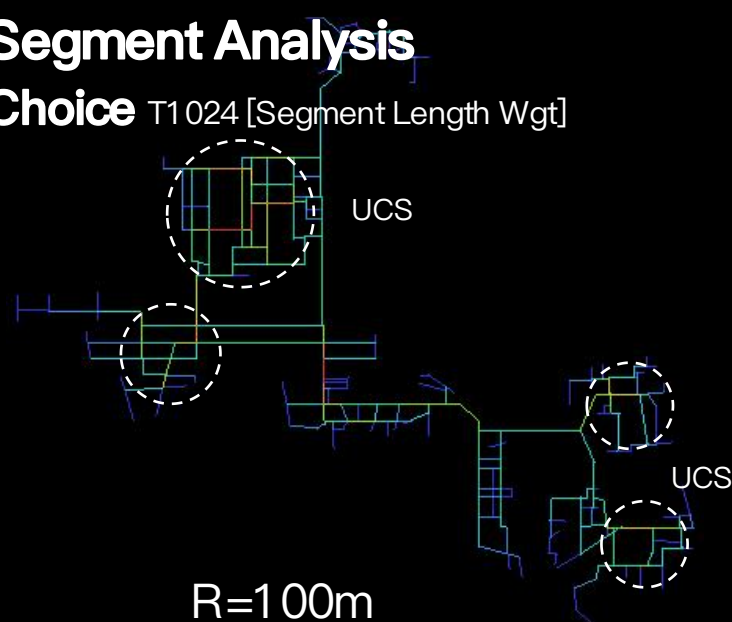
Areas with higher integration values exhibit greater spatial accessibility.

As the radius (R) increases to the global scale, the main underground corridors show stronger integration.

3. Interior Space Characteristics of SJTUUS

Segment Analysis

Choice T1024 [Segment Length Wgt]



Characteristics of High Choice Areas:

$R = 100\text{ m}$: mainly underground commercial spaces.

$R = 200\text{ m}$: underground commercial spaces+secondary corridors.

$R = 300\text{ m}$: shift toward the main corridors.

$R = 500\text{ m} / n$: fully concentrated along the main corridors.

Conclusion:

$R = 200\text{ m}$ serves as the threshold point.

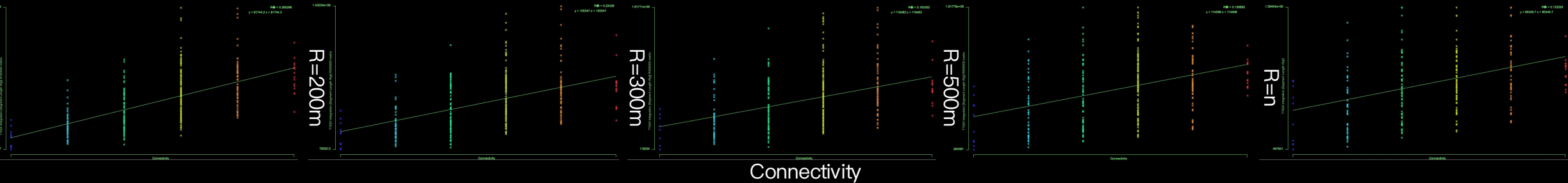
When $R < 200\text{ m}$, the center of choice is focused on underground commercial spaces.

When $R > 200\text{ m}$, the center shifts toward the main underground corridors.

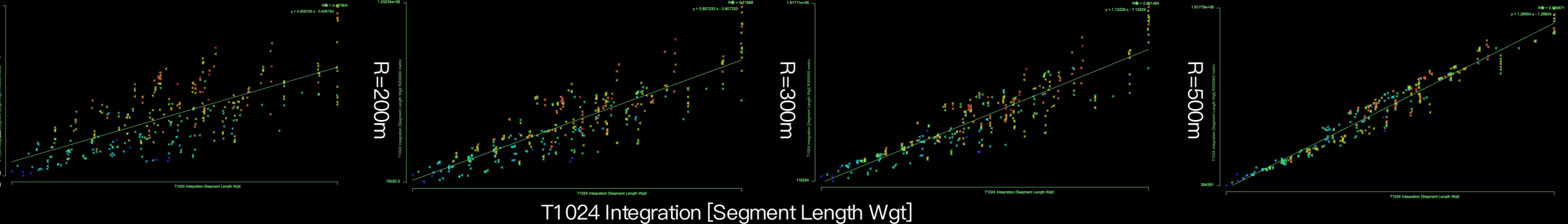
3. Interior Space Characteristics of SJTUUS

Segment Analysis

Intelligibility degree: the correlation coefficient of the integration degree and the connectivity value



Synergy degree: by the correlation coefficient of the global integration degree and the local integration degree



R ²	R=100 m	R=200 m	R=300 m	R=500 m	R=n
Intelligibility	0.49	0.72	0.80	0.96	—
Synergy	0.37	0.22	0.16	0.14	0.1 3

- Intelligibility Characteristics:

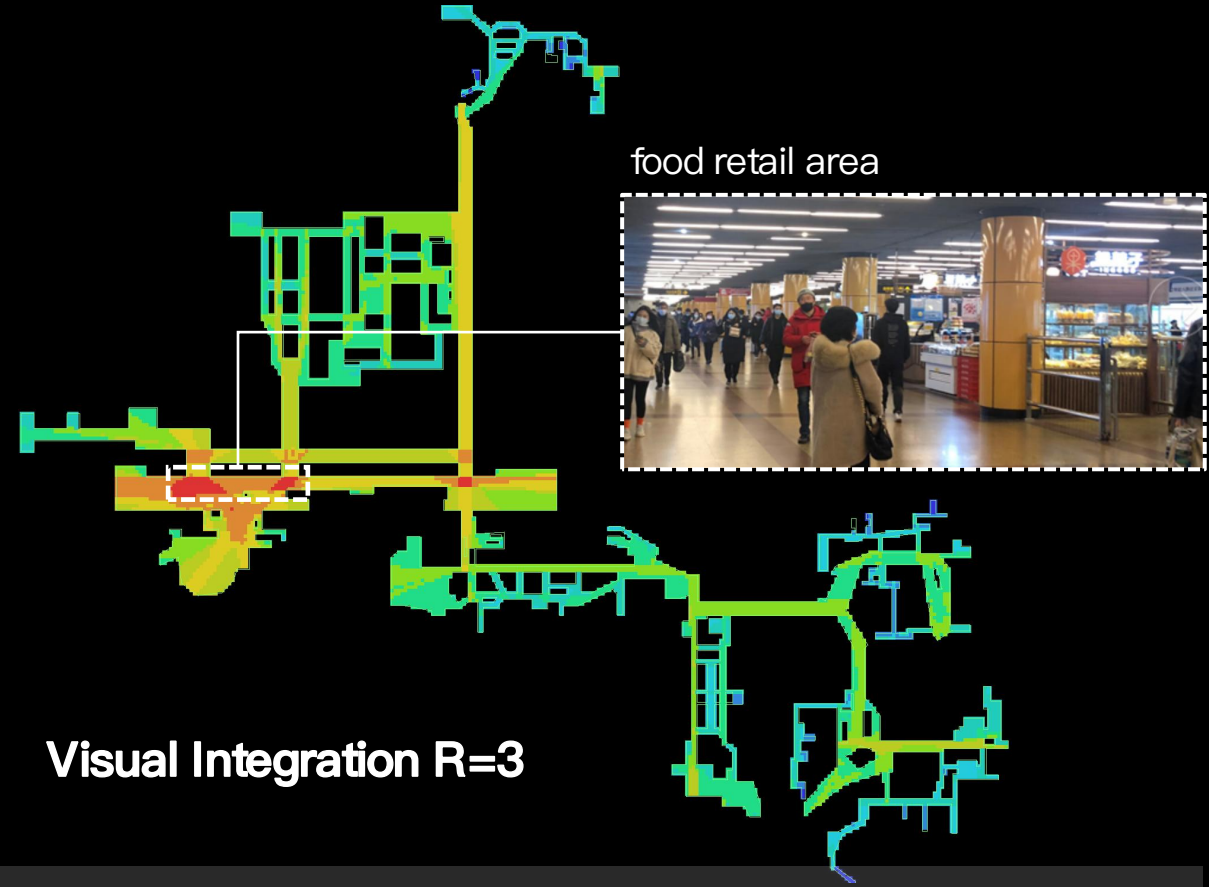
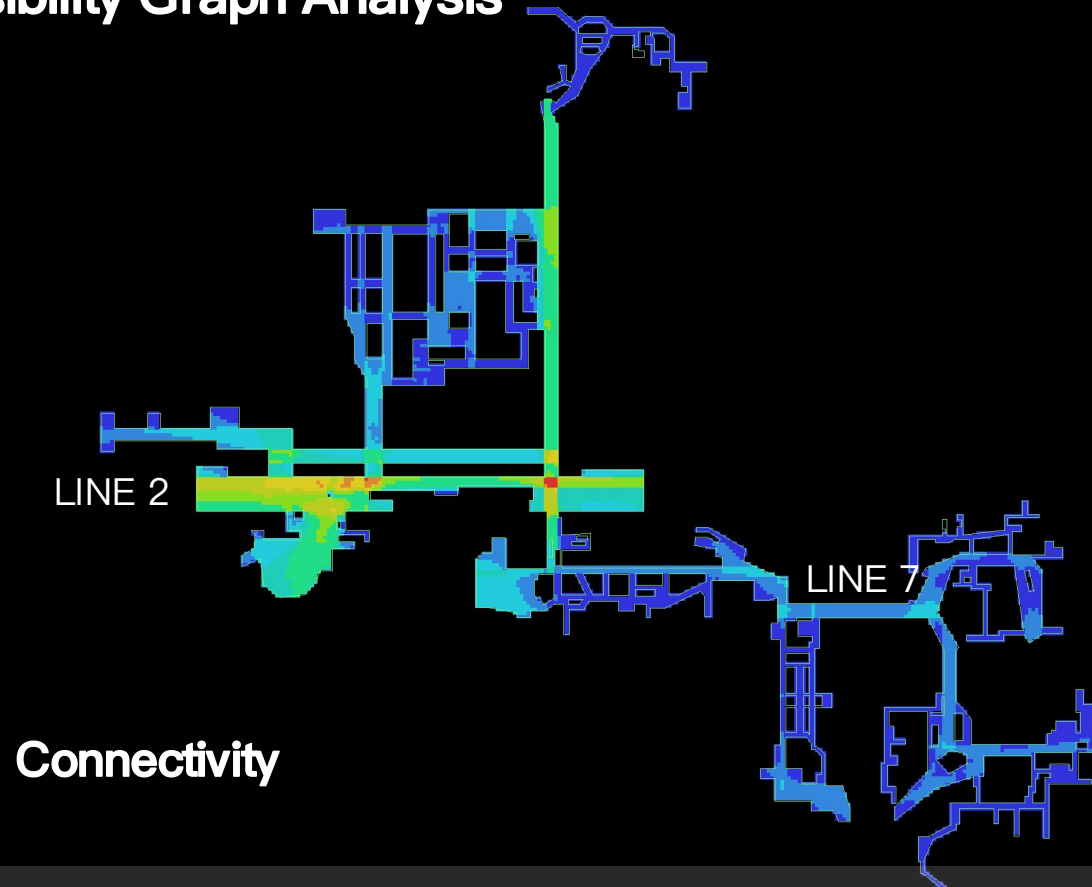
When **R equals 100 meters**, the integration value is below 0.5, which means the space is mostly **localized and independent** — in this case, the **underground commercial spaces** (UCS). People standing in these **areas can’ t easily understand or perceive the overall structure** of the underground space.

Synergy Characteristics:

The low R² value indicates a **low level of overall spatial perception**, which is **unfavorable for enhancing spatial vitality**.

3. Interior Space Characteristics of SJTUUS

Visibility Graph Analysis

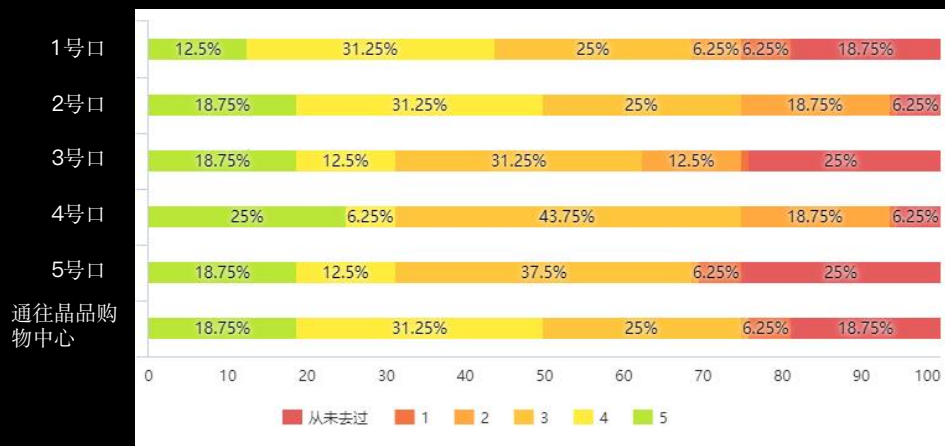


- The visibility of UPS is higher, and the visibility of metro corridors is higher than that of commercial corridors.
- Within the UPS, the food vending area has the highest visibility, and this layout pattern effectively enhances commercial value and attracts pedestrian flow.
- The visibility of Line 2's station concourse is higher than that of Line 7
—— consistent with the following survey results (spatial satisfaction is also higher in Line 2 area)

3. Interior Space Characteristics of SJTUUS

➤ Pedestrian Corridor Space Satisfaction Survey

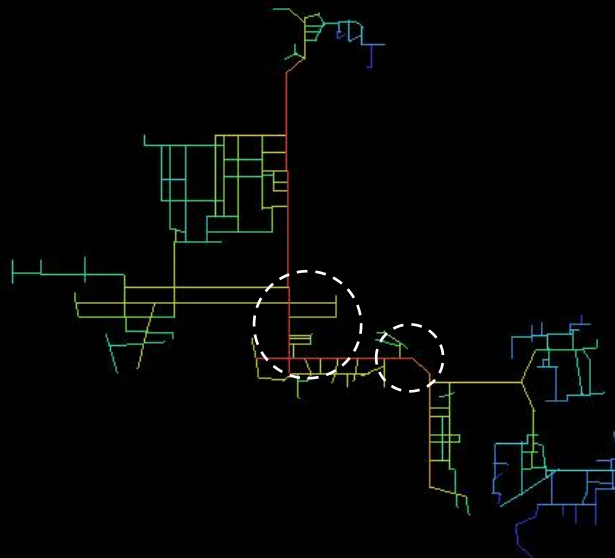
Line 2 Questionnaire Results



Line 7 Questionnaire Results



Integration



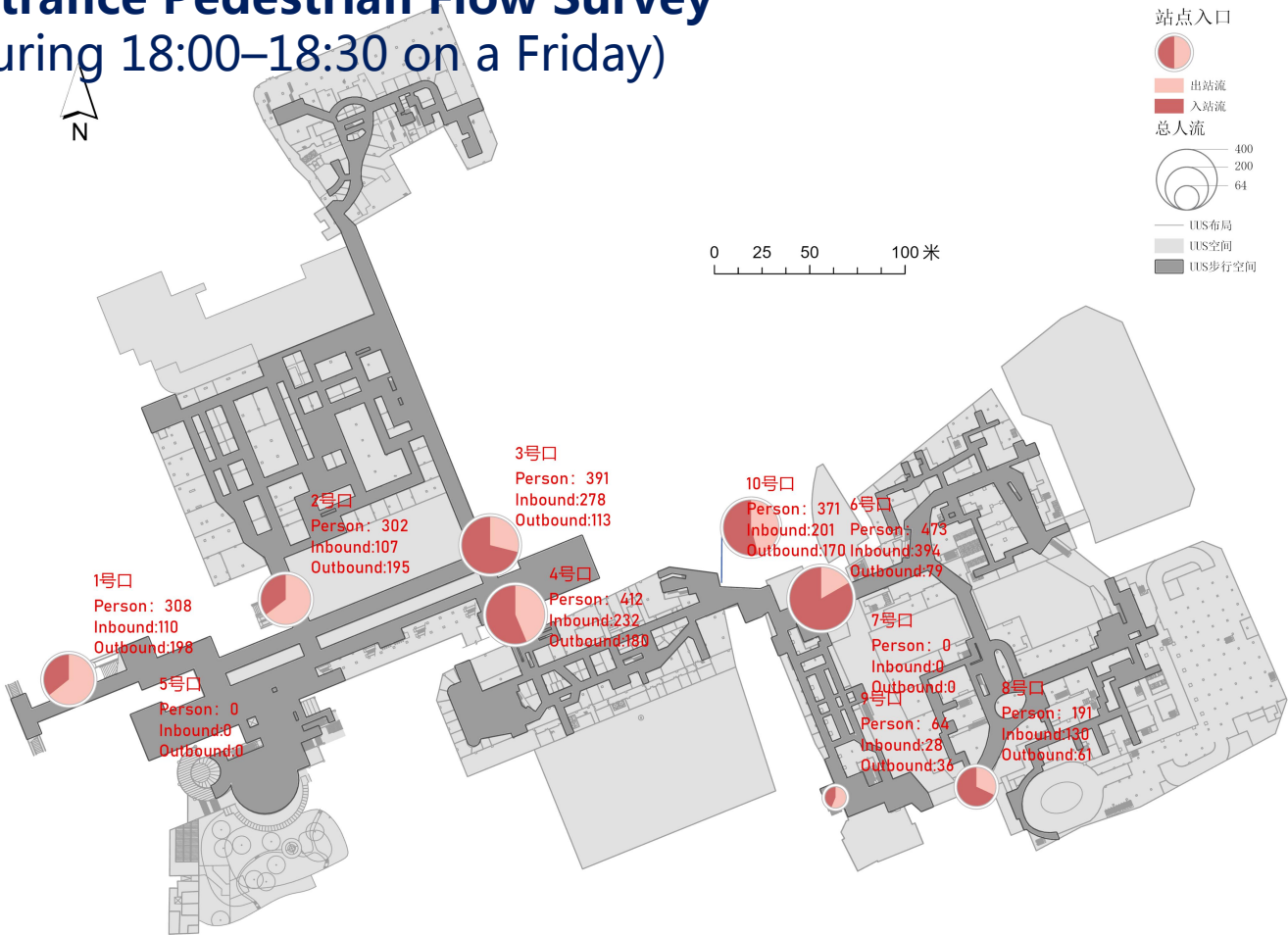
Visual Integration R=3



- In terms of **spatial perception**, the overall cognition level at **Line 2 entrances** is higher than that of Line 7, which may be related to its higher accessibility and visibility.
- **Entrances 4 and 10** show stronger spatial perception, which may also be attributed to their higher accessibility and visibility.

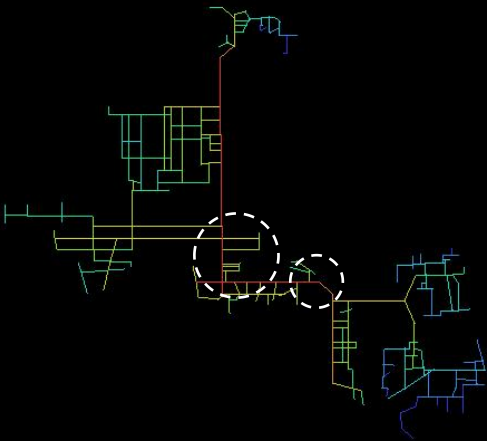
3. Interior Space Characteristics of SJTUUS

Entrance Pedestrian Flow Survey (during 18:00–18:30 on a Friday)

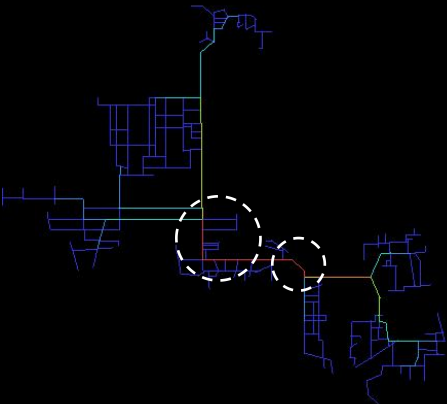


	1号口	2号口	3号口	4号口	6号口	8号口	9号口	10号口	5号口	7号口	总计
进	198	107	278	232	394	130	28	201	0	0	1568
出	110	195	113	180	79	61	36	170	0	0	944
总计	308	302	391	412	473	191	64	371	0	0	2512

Integration



Choice



Analysis Results:

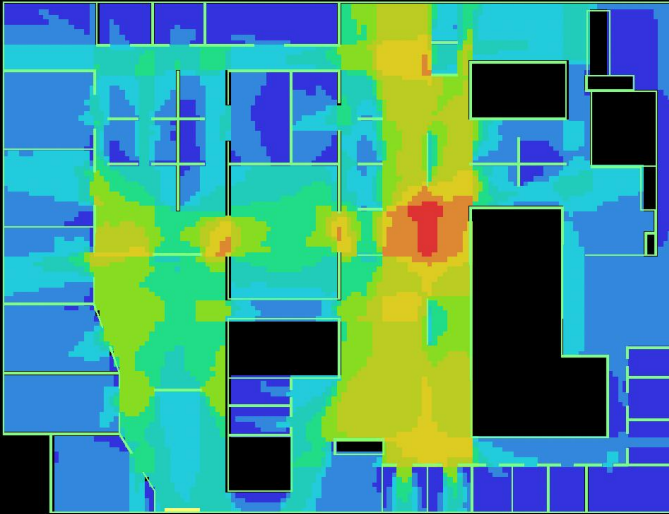
- On-site pedestrian flow surveys indicate that these three entrances (Entrances 3, 4, and 6) experienced the highest pedestrian.
- The UPS near these three entrances shows higher spatial choice and integration values.

the spatial efficiency around these entrances is relatively high.

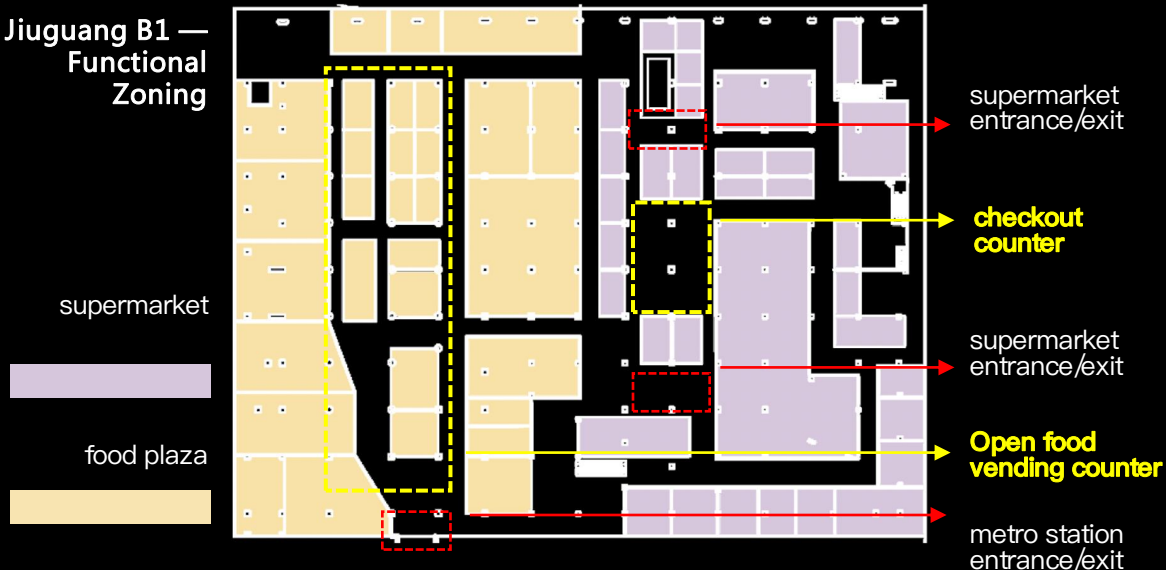
3. Interior Space Characteristics of SJTUUS — UCS

Underground Commercial Space (UCS) — **Jiuguang Department Store** as a Case Study

VGA Visibility



Jiuguang B1 —
Functional
Zoning

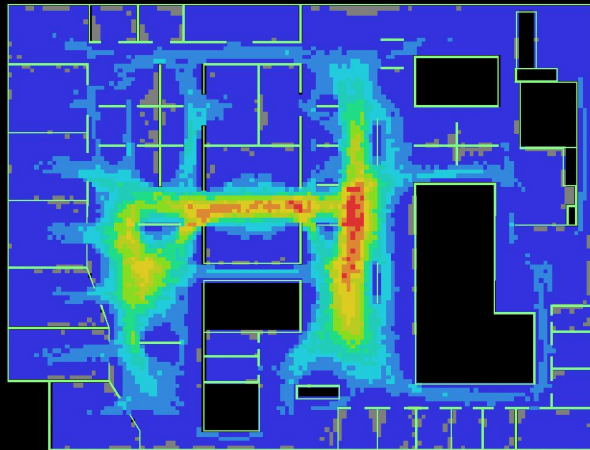


Overall VGA Visibility Analysis Results:

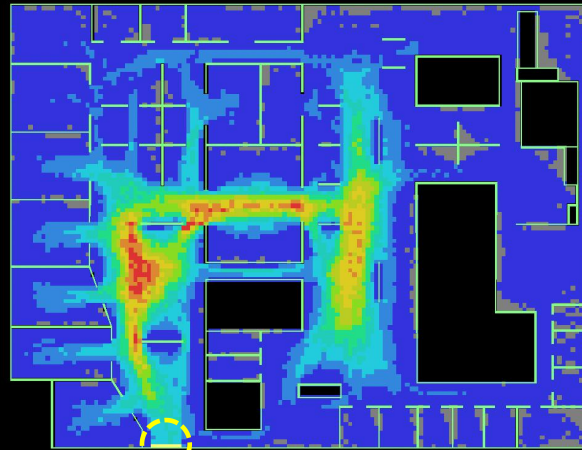
- The **checkout counter area** inside the supermarket shows the highest visibility, which facilitates crowd dispersal and efficient checkout, while also attracting customers deeper into the supermarket.
- The **open food vending counter** also demonstrates the highest visibility. This layout effectively draws the attention of passing customers, encourages impulse purchases, and enhances commercial value.

3. Interior Space Characteristics of SJTUUS — UCS

Underground Commercial Space (UCS) — **Jiuguang Department Store** as a Case Study

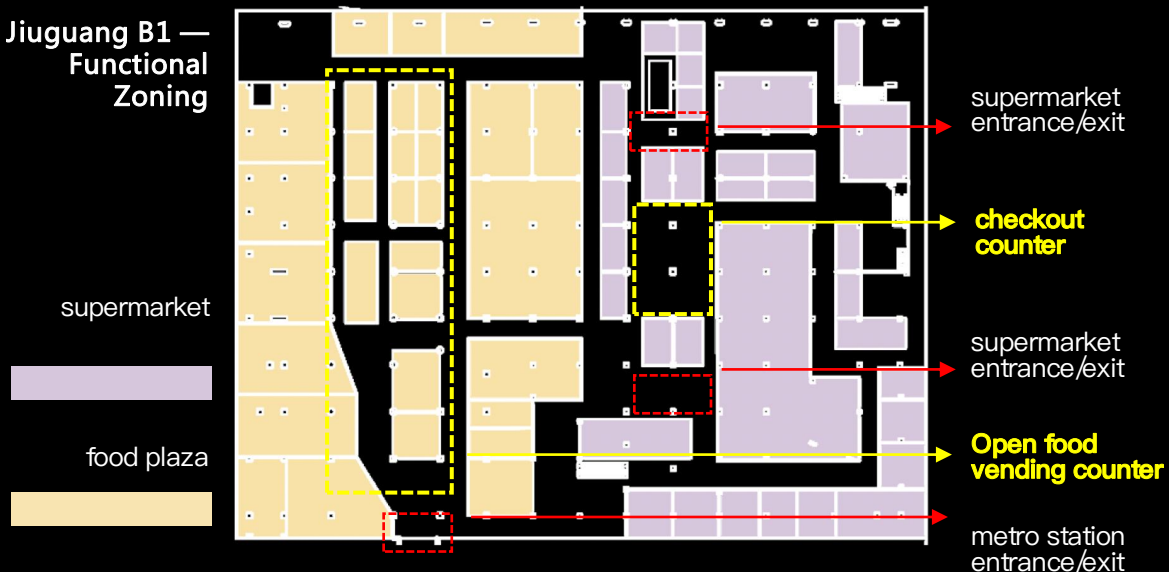


Agent Behavior — Random Point Distribution



Agent Behavior — Entrance Point Distribution

Jiuguang B1 —
Functional
Zoning



Agent Behavior Analysis Results:

- Under global random point distribution

agent trajectories were mainly concentrated around the checkout area and open-style food vending zones. This layout not only **enables higher economic returns, but also helps supermarket customers check out quickly and facilitates efficient crowd dispersal.**

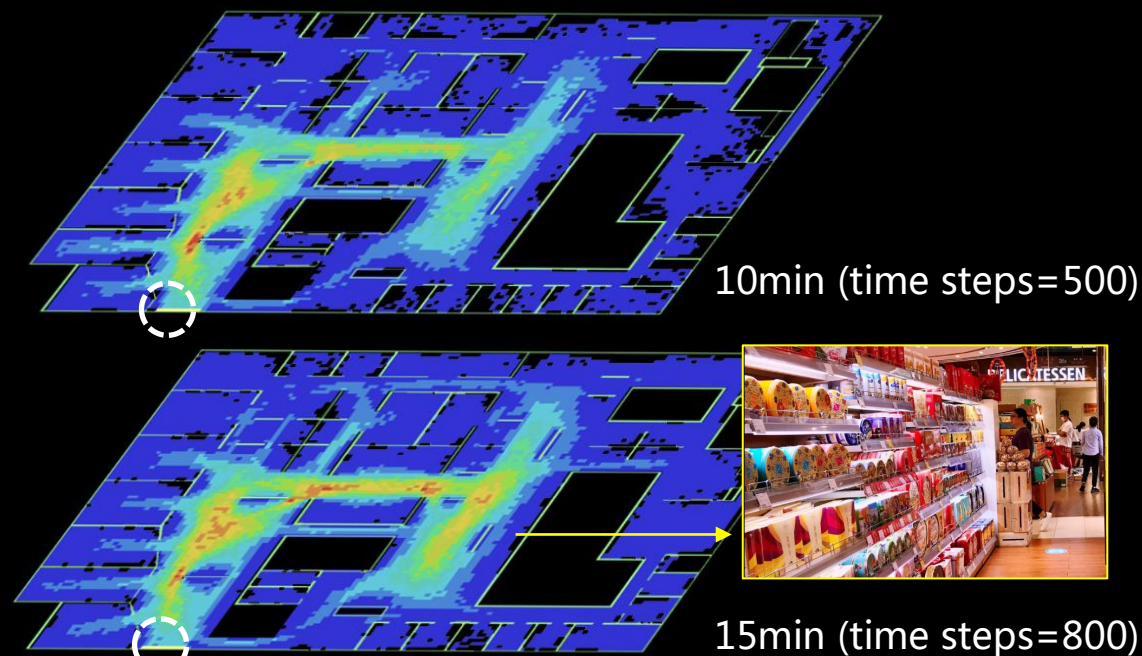
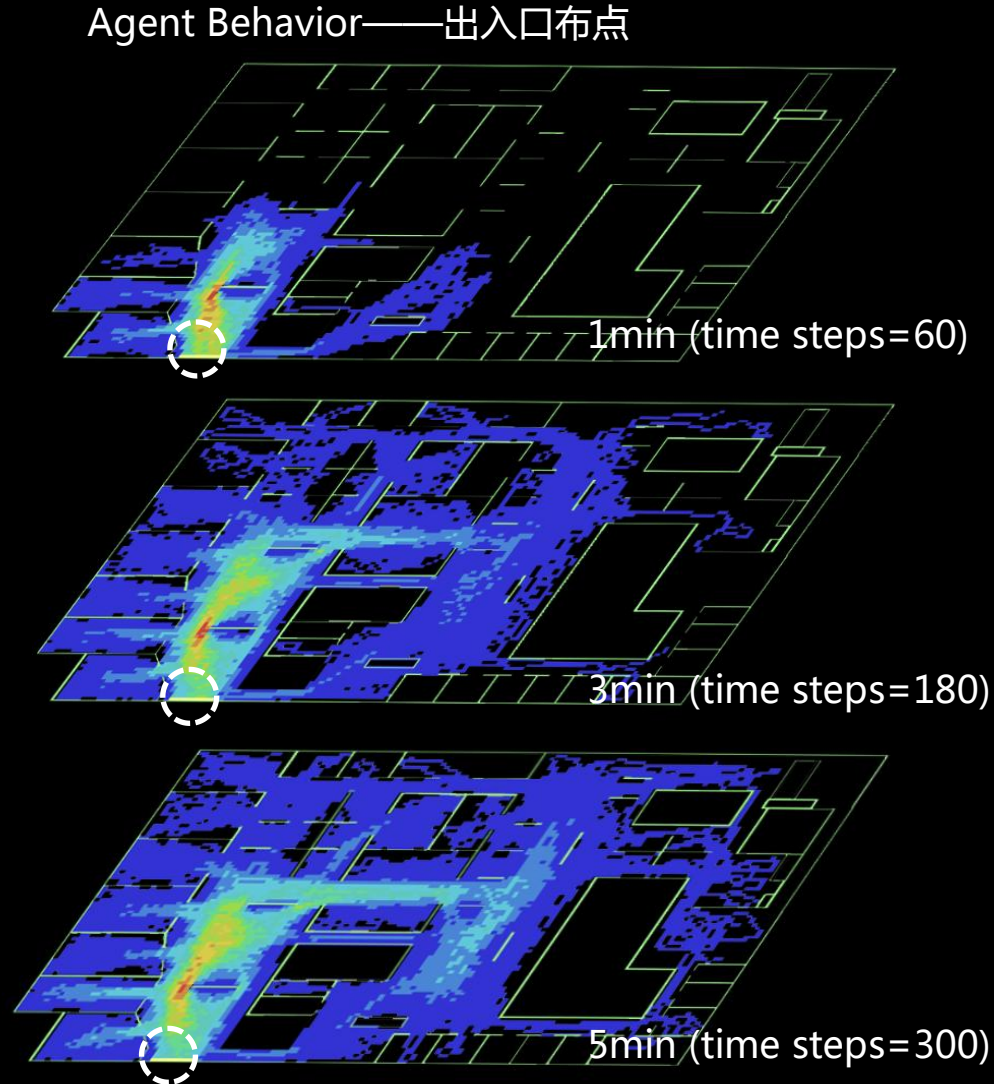
- Under fixed-point observation at metro station entrances

global agent trajectories showed **significant aggregation in the open food vending areas.** The passing flow plays a key role in activating these zones, indirectly demonstrating the rationality of this layout model, which can **effectively attract pedestrian attention and increase economic benefits.**

3. Interior Space Characteristics of SJTUUS — UCS

Underground Commercial Space (UCS) — Jiuguang Department Store as a Case Study

Agent Behavior——出入口布点



□ Agent Behavior Analysis Results:

As walking time increased, the checkout areas and entrances of the supermarket also became major pedestrian convergence points

- verifies the rationality of the mall layout from the perspective of maximizing economic benefits
- effectively attracts pedestrian flows into the supermarket, and the snack shelves near checkout areas further encourage quick purchases and rapid flow turnover.

Space syntax provides a quantitative approach to analyzing the spatial structure and characteristics of Urban Underground Space (UUS). By measuring key indicators such as accessibility, visibility, and intelligibility of both external and internal spaces of SJTUUS, and combining these with site analysis at two spatial scales, this study **identifies the relationship between spatial configuration and functional distribution,** and **proposes strategies to optimize UUS spatial efficiency through spatial adjustments.**

- ✓ The number and layout of station entrances should be coordinated with surrounding land use and POI distribution, and located along high-accessibility streets to better match passenger flow demand.

Underground Public Space (UPS)

- ✓ To improve users' understanding of the overall space, **the spatial structure should be optimized to enhance intelligibility**, thereby stimulating spatial vitality, especially at key entrances with high pedestrian flows.
- ✓ Enhancing **the visibility of concourses and main corridors** can **improve the user experience**, while increasing the visibility of underground commercial streets can also boost economic performance.

Underground Commercial Space (UPS)

- ✓ **Fast consumption area** such as open-style food vending areas, snack shelves, and **high-flow nodes** such as checkout counters **should be located in areas with high visibility** to improve both economic benefits and space utilization efficiency.
- ✓ **Functional layout planning should integrate spatial functions with pedestrian flow directions** to **maximize overall efficiency and benefits.**